

Institución: UPY

Materia: Programación (Segundo Cuatrimestre ° Cuatrimestre)

Profesor: Jorge Alejandro Pavia Rivero

Alumno: Sayid Israel Saadi Gutiérrez | **Grupo:** 2A

Fecha: 2026-07-11

HW 2 - U2 - Selective and Iterative Control Structures

```
# Problem 1 - Grade Averaging System
# HW 2 - U2 - Selective and Iterative Control Structures

# INPUT
total = 0.0          # accumulator for the sum of grades
count = 0            # counter of grades entered

entry = input("Enter a grade (or 'done' to finish): ")

# PROCESS
while entry != "done":
    grade = float(entry)
    total = total + grade
    count = count + 1
    entry = input("Enter a grade (or 'done' to finish): ")

# OUTPUT
if count == 0:
    print("No grades entered. Please enter at least one grade.")
else:
    average = total / count
    if average >= 7.0:
        print(f"Average: {average:.2f} – Passed")
    else:
        print(f"Average: {average:.2f} – Failed")
```

Problem 1 — Grade Averaging System

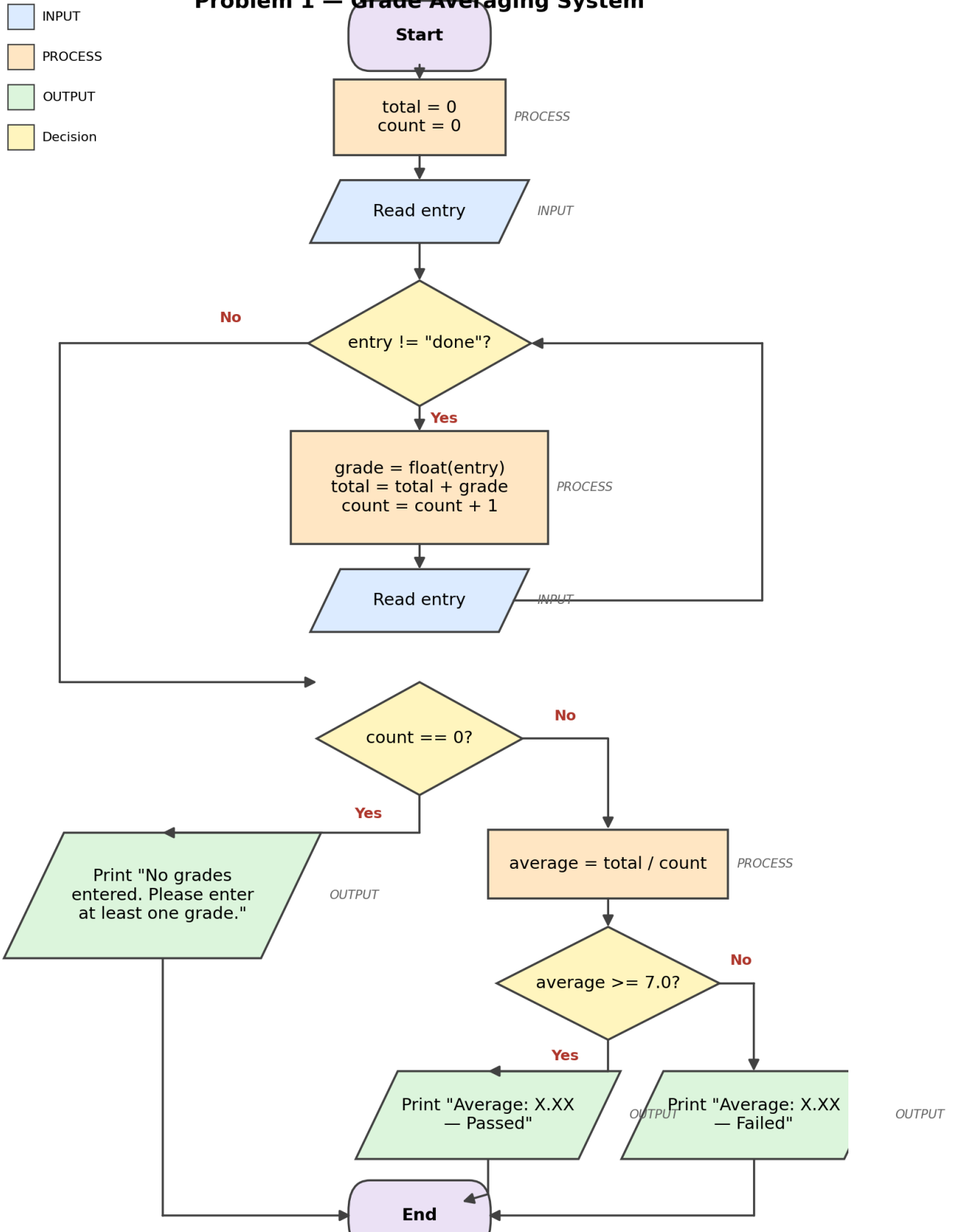


Figura 1:

```
# Problem 2 - BMI Calculator
# HW 2 - U2 - Selective and Iterative Control Structures

# INPUT
entry = input("Enter weight in kg (or 'exit' to finish): ")

while entry != "exit":
    weight = float(entry)
    height = float(input("Enter height in m: "))

    # PROCESS
    bmi = weight / height ** 2

    if bmi < 18.5:
        category = "Underweight"
    elif bmi < 25:
        category = "Normal"
    elif bmi < 30:
        category = "Overweight"
    else:
        category = "Obese"

    # OUTPUT
    print(f"BMI: {bmi:.2f} – {category}")

    entry = input("Enter weight in kg (or 'exit' to finish): ")
```

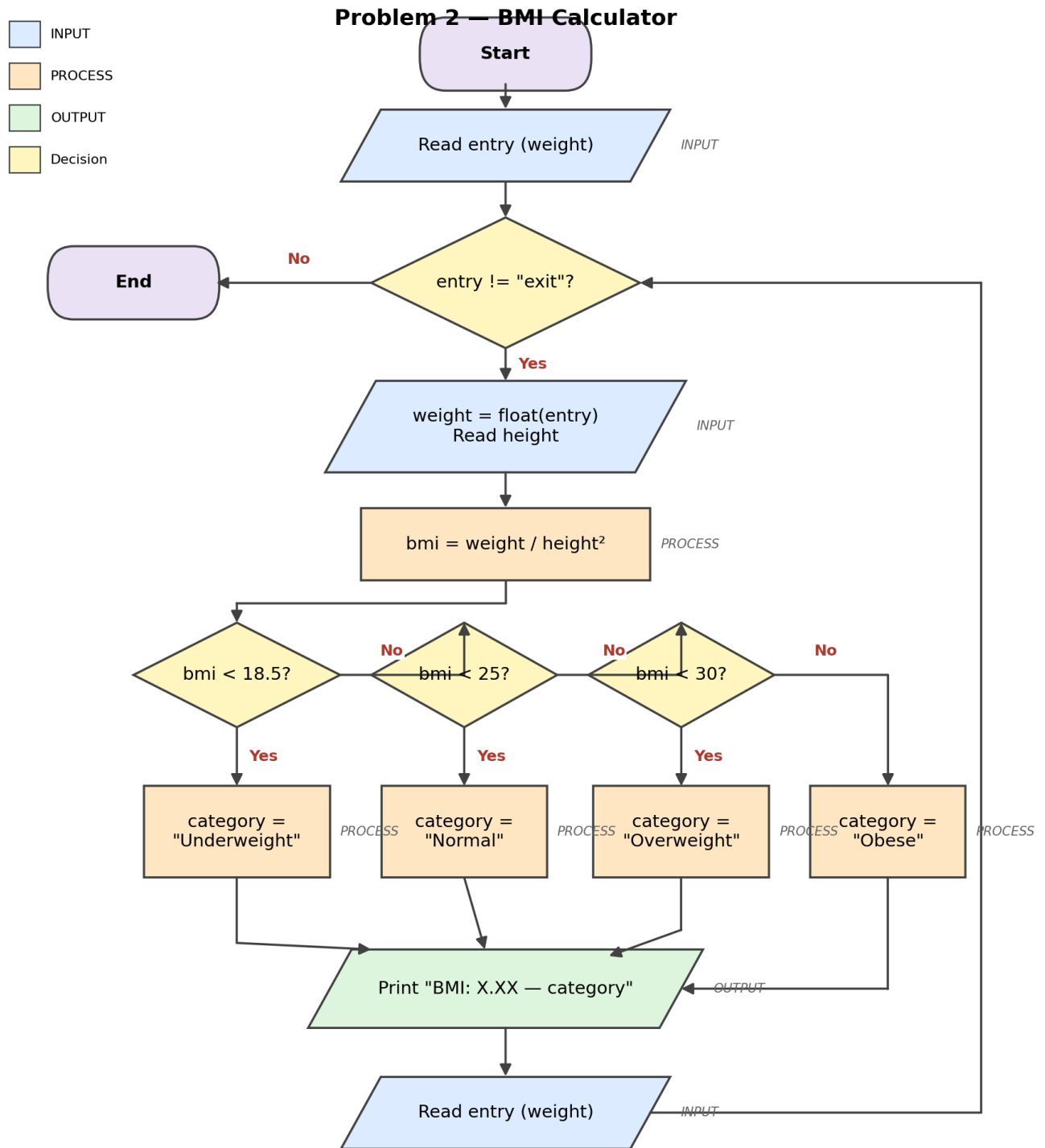


Figura 2:

```
# Problem 3 - Water Bill Calculator
# HW 2 - U2 - Selective and Iterative Control Structures

# INPUT
total = 0.0      # accumulator for the total charge

entry = input("Enter m3 consumed (or 'exit' to finish): ")

while entry != "exit":
    consumption = float(entry)

    # PROCESS
    if consumption <= 10:
        rate = 8.00
    elif consumption <= 20:
        rate = 12.00
    else:
        rate = 18.00

    charge = consumption * rate
    total = total + charge

    # OUTPUT
    print(f"Month charge: ${charge:.2f} MXN")

    entry = input("Enter m3 consumed (or 'exit' to finish): ")

print(f"Total: ${total:.2f} MXN")
```

Problem 3 — Water Bill Calculator

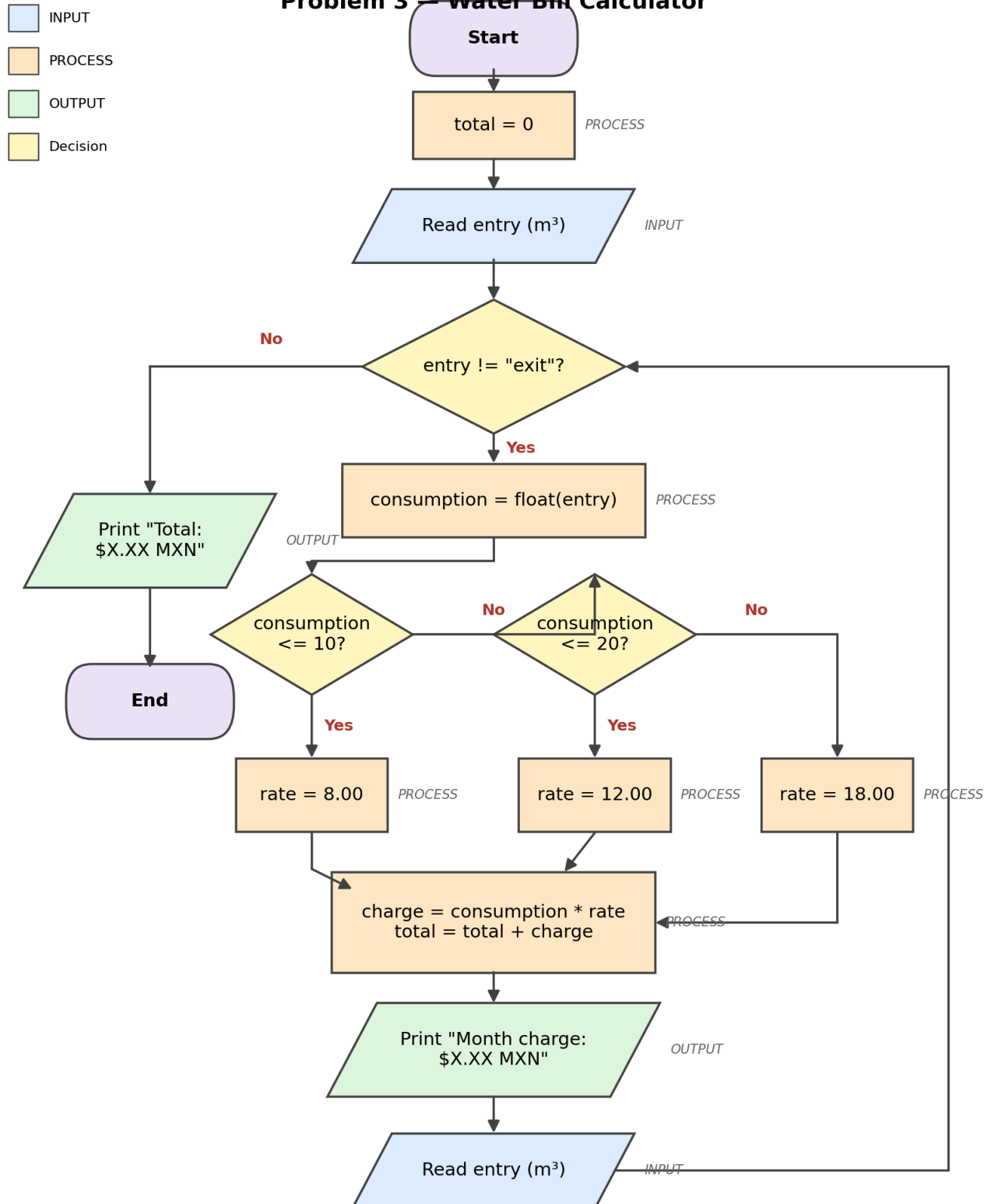


Figura 3:

```
# Problem 4 - Season Classifier
# HW 2 - U2 - Selective and Iterative Control Structures

# INPUT
entry = input("Enter a month number 1-12 (or 'exit' to finish): ")

while entry != "exit":
    month = int(entry)

    # PROCESS
    if month < 1 or month > 12:
        result = "Invalid month. Please enter a number between 1 and 12."
    elif month == 12 or month == 1 or month == 2:
        result = "Winter"
    elif month >= 3 and month <= 5:
        result = "Spring"
    elif month >= 6 and month <= 8:
        result = "Summer"
    else:
        result = "Fall"

    # OUTPUT
    print(result)

    entry = input("Enter a month number 1-12 (or 'exit' to finish): ")
```

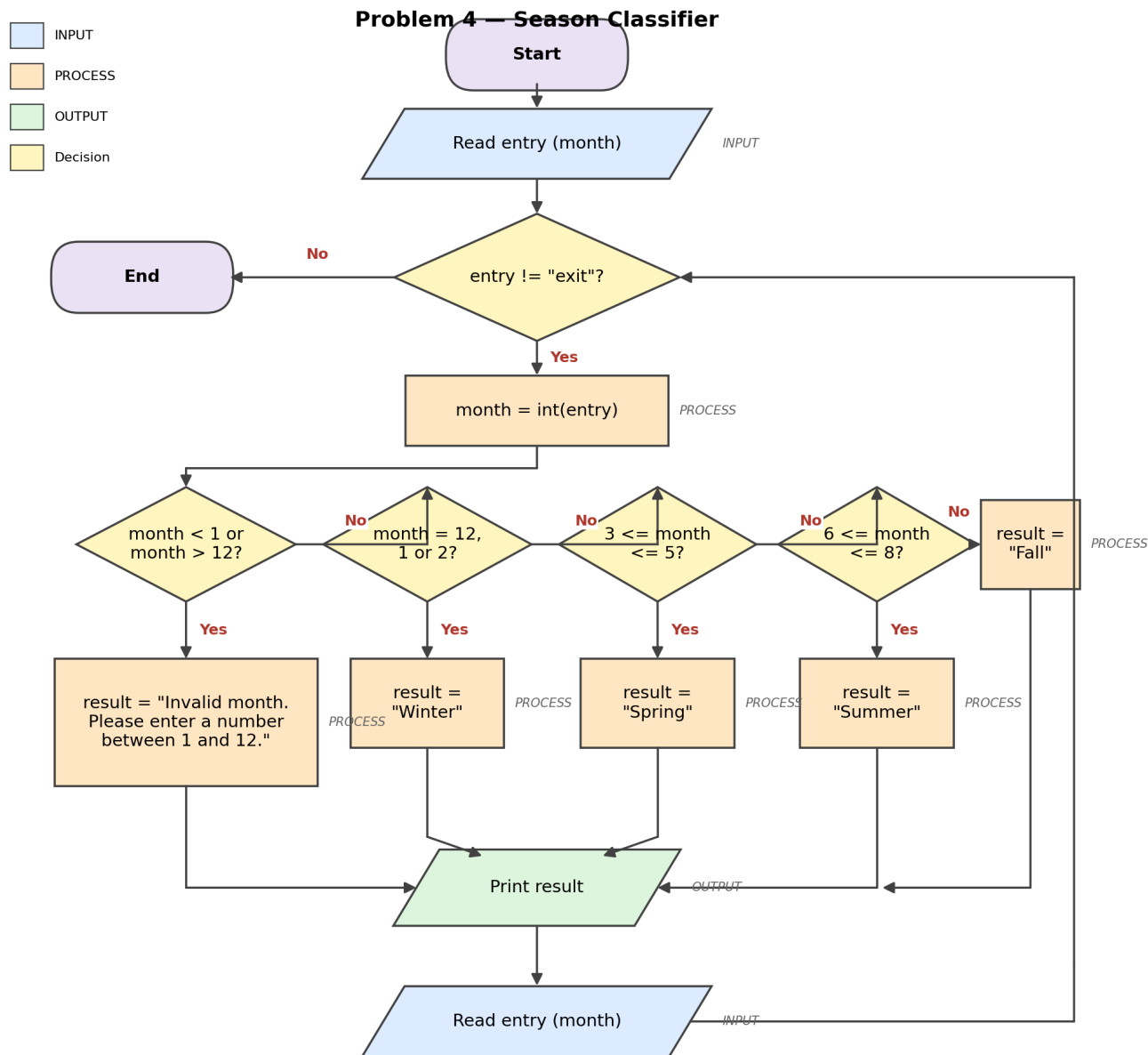



Figura 4:

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# Problem 5 - Shipping Cost Calculator
# HW 2 - U2 - Selective and Iterative Control Structures
#
# Rate table:
#
#           weight <= 5 kg   weight > 5 kg
# distance <= 100 km   $50.00 MXN   $80.00 MXN
# distance > 100 km   $120.00 MXN   $200.00 MXN

# INPUT
total = 0.0      # accumulator for the total shipping cost

entry = input("Enter package weight in kg (or 'exit' to finish): ")

while entry != "exit":
    weight = float(entry)
    distance = float(input("Enter distance in km: "))

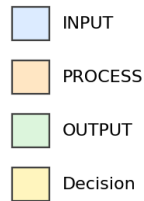
    # PROCESS
    if distance <= 100:
        if weight <= 5:
            cost = 50.00
        else:
            cost = 80.00
    else:
        if weight <= 5:
            cost = 120.00
        else:
            cost = 200.00

    total = total + cost

# OUTPUT
print(f"Shipping cost: ${cost:.2f} MXN")

```

```
entry = input("Enter package weight in kg (or 'exit' to finish): ")  
  
print(f"Total: ${total:.2f} MXN")
```



Problem 5 – Shipping Cost Calculator

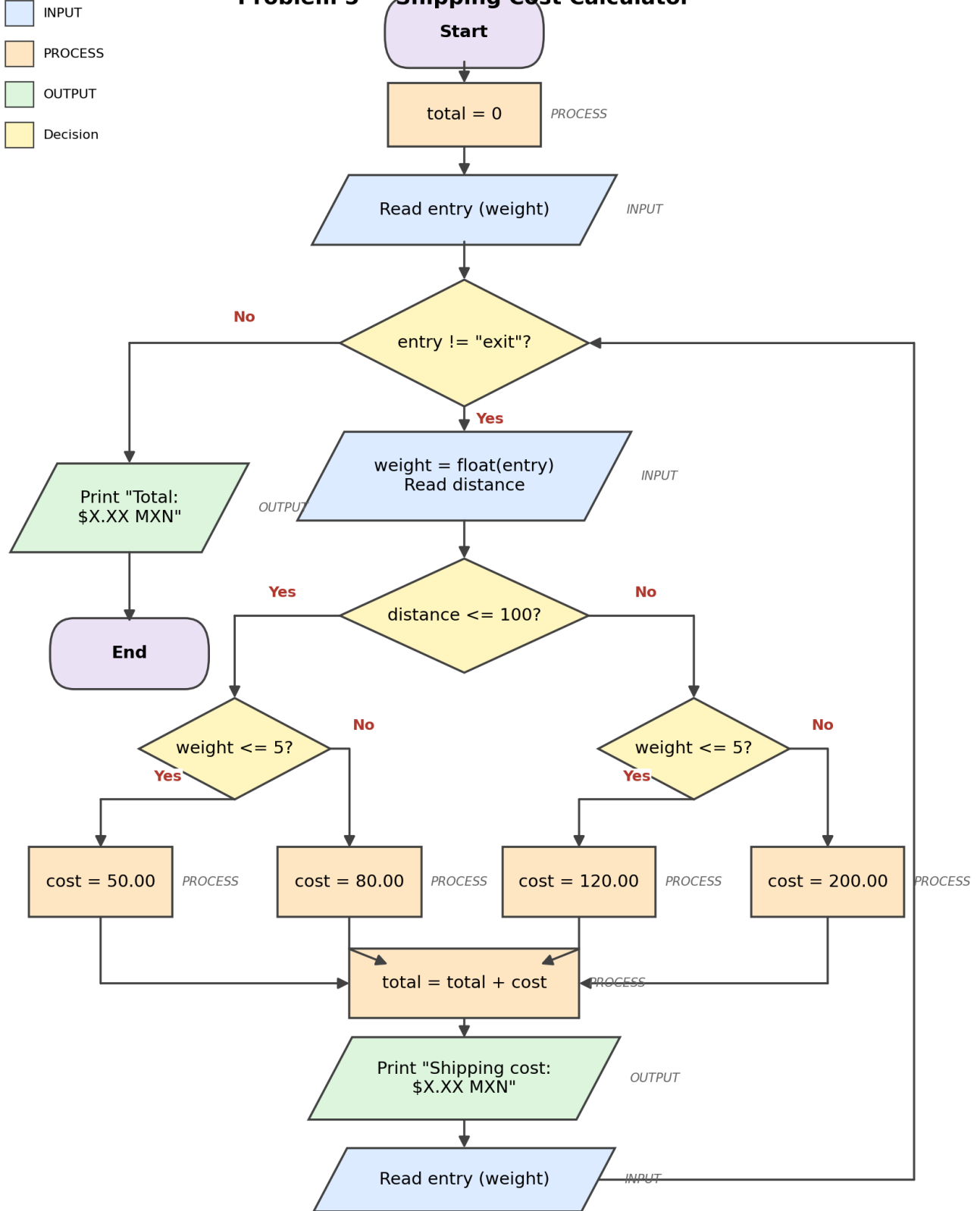


Figura 5:

Nombre del estudiante: Sayid Israel Saadi Gutiérrez

IA utilizada: Claude

Fecha de uso: 2026-07-11

Propósito: Verificación de código y creación de diagramas

Prompt utilizado:

Verifica qué mis códigos sean correctos y crea las bases de los diagramas correspondientes

Respuesta en crudo (raw):